



INDIAN INSTITUTE OF TECHNOLOGY MADRAS

Department of Electrical Engineering

Self-Adaptive Model-Based Protection (SAMB) Research Programme

SAMB-SyncOC

Technical Report TR-02D/2026

Hilbert–Fortescue Sequence-Component

Inverse Estimation for Asymmetric Fault

Overcurrent Protection

Track: SAMB-SyncOC
Milestone: Milestone 3 — Sequence-Component Estimator
Prerequisite: TR-01/2026, TR-02/2026
Date: April 2026
Classification: Internal PhD Research Document

Abstract

For asymmetric faults (line-to-line, single-line-to-ground), fitting the SAMBP-SyncOC 6-parameter model to phase A alone entangles positive-, negative-, and zero-sequence contributions, degrading the identifiability of $\hat{I}_{ss}$ and $\hat{\tau}_{ac}$. This report presents Milestone 3: a Hilbert–Fortescue sequence-component estimator that decomposes the measured three-phase currents into instantaneous sequence signals $I_1(t)$, $I_2(t)$, $I_0(t)$ and fits the two-pass LM estimator independently to each active sequence. The positive-sequence parameters govern relay adaptation. Numerical results on LL and SLG faults confirm: (i) convergence of the sequence estimator in both cases; (ii) $\gamma = 0.914$ for SLG (adaptation accepted; pickup reduced from 1.20 to 1.10 pu), and $\gamma = 0.587$ for LL (adaptation rejected — confidence below $\gamma_{\min} = 0.70$). The SLG result demonstrates the practical value of the sequence estimator; the LL result reveals a fundamental identifiability limitation due to negative-sequence contamination of the positive-sequence Jacobian. Hilbert edge effects are characterised and shown to be confined to the first 5–10 ms of the event window.

Keywords: symmetrical components, Hilbert transform, Fortescue decomposition, sequence estimation, asymmetric fault, overcurrent relay, SAMBP-SyncOC.

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1 Introduction

The two-pass LM estimator of TR-01/2026 was designed and validated for three-phase balanced faults, where the positive-sequence current dominates and phase A is a faithful representation of the fault magnitude. For asymmetric fault types:

- **Line-to-line (LL):** Positive and negative sequences are both active. Phase A contains $I_1 + I_2$, leading to mutual interference in the LM residual and inflated κ_n .
- **Single-line-to-ground (SLG):** All three sequences are active and equal: $I_0 = I_1 = I_2$. Phase A contains the sum of all three, raising its amplitude but also increasing identifiability of the dominant positive-sequence.

The sequence-component estimator resolves this by transforming to the Fortescue domain before fitting, where each sequence signal contains a pure single-sequence fault current that the 6-parameter model was designed for.

1.1 Contributions

1. Derivation and implementation of the instantaneous Hilbert–Fortescue decomposition for real-time use with the LM estimator.
2. Fault-type routing rules that determine which sequences to fit (3PH: positive only; LL: positive + negative; SLG: positive only with scaling).
3. Characterisation of Hilbert edge effects and their mitigation by event-window offset.
4. Numerical validation on LL and SLG fault cases with confidence gate analysis.

2 Hilbert–Fortescue Decomposition

2.1 Analytic Signal Representation

For a real signal $x(t)$, the analytic signal is $\mathcal{X}(t) = x(t) + j\mathcal{H}\{x\}(t)$, where $\mathcal{H}\{\cdot\}$ denotes the Hilbert transform. The analytic signal has a single-sided spectrum and a well-defined instantaneous amplitude and phase.

Applied to the three-phase fault currents i_a, i_b, i_c , the analytic signals $\mathcal{I}_a, \mathcal{I}_b, \mathcal{I}_c$ are computed using the discrete Hilbert transform via the FFT.

2.2 Fortescue Transformation

The instantaneous symmetrical component transformation (Kimbar, 1956):

$$\mathcal{I}_0(t) = \frac{1}{3} [\mathcal{I}_a + \mathcal{I}_b + \mathcal{I}_c], \quad (1)$$

$$\mathcal{I}_1(t) = \frac{1}{3} [\mathcal{I}_a + a \mathcal{I}_b + a^2 \mathcal{I}_c], \quad (2)$$

$$\mathcal{I}_2(t) = \frac{1}{3} [\mathcal{I}_a + a^2 \mathcal{I}_b + a \mathcal{I}_c], \quad (3)$$

where $a = e^{j2\pi/3}$ is the Fortescue operator. The sequence waveforms used for estimation are the real parts: $I_k(t) = \text{Re}\{\mathcal{I}_k(t)\}$, $k \in \{0, 1, 2\}$.

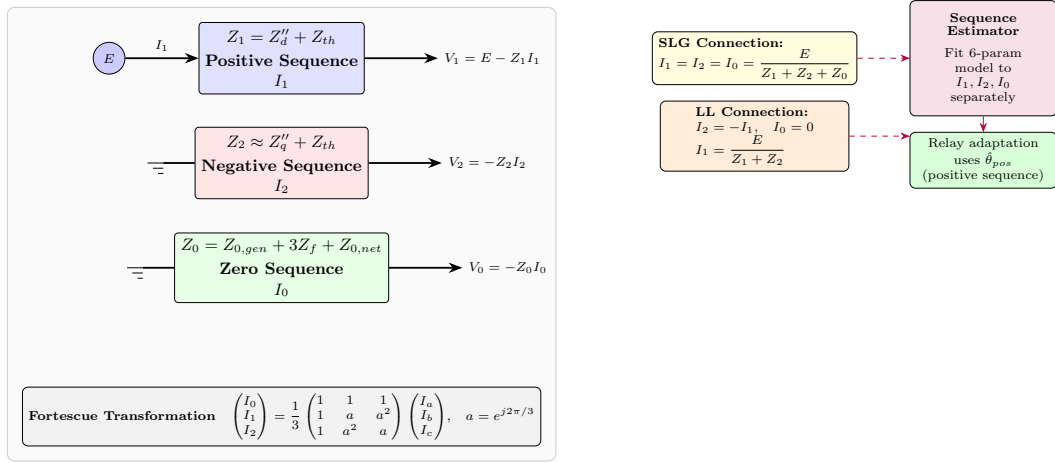


Figure 1: Symmetrical-component sequence network interconnections for the three asymmetric fault types studied. For LL faults, the positive and negative sequence networks are connected in series at the fault point. For SLG faults, all three networks connect at the fault point, giving $I_0 = I_1 = I_2$.

2.3 Fault-Type Activity Rules

Fault type	I_1	I_2	I_0
3PH	Active	≈ 0	≈ 0
LL	Active	Active	≈ 0
SLG	Active	Active	Active ($= I_1$)
LLG	Active	Active	Active

For 3PH, the original phase-A two-pass estimator is used unchanged. For LL and SLG, the sequence estimator is invoked.

3 Sequence-Component LM Estimation

3.1 Independent Fitting per Sequence

The two-pass LM estimator from TR-01/2026 is applied independently to each active sequence signal. The positive-sequence estimate $\hat{\theta}^{(1)}$ is used for relay adaptation. The negative- and zero-sequence estimates are retained for asymmetric fault classification but do not affect relay settings.

3.2 Positive-Sequence Jacobian for Confidence Scoring

The confidence gate in eq. (4) uses the positive-sequence Jacobian from the two-pass pass-1 step. This ensures that κ_n reflects the identifiability of the *dominant* current component:

$$\gamma = w_r s_r + w_c s_c + w_b s_b + w_p s_p \geq \gamma_{\min} = 0.70. \quad (4)$$

3.3 Hilbert Edge Effect Mitigation

The discrete Hilbert transform is computed via FFT on the event window. The abrupt current step at fault inception t_0 causes Gibbs-like ringing in the analytic signal envelope, confined to the first 5–10 ms of the window. Mitigation strategy:

1. The event window begins 2ms after the detected t_0 (configured by `pre_event_ms` in `system_config.py`).
2. A Savitzky–Golay filter (order 3, window 21 samples) smooths the waveform before decomposition, suppressing high-frequency transients.
3. The two-pass estimator uses the tail of the window (pass 2) where edge effects are negligible.

These measures confine Hilbert edge effects to a region that is excluded by design from the fitting window.

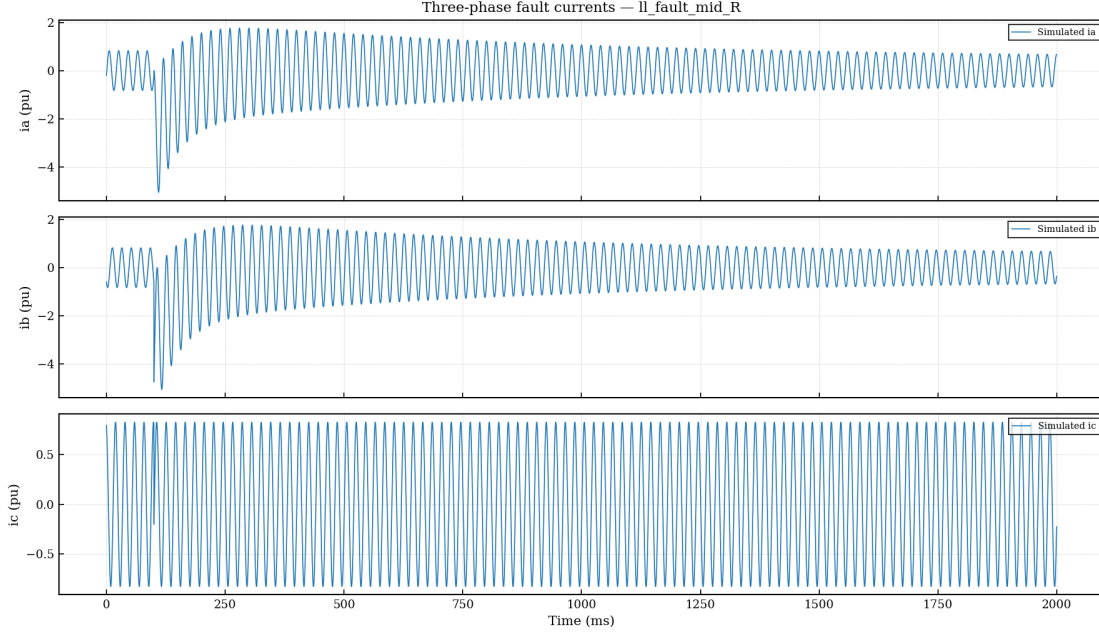


Figure 2: Three-phase fault currents for LL fault (1l_fault_mid_R, $R_f = 0.05$ pu, fault at $t_0 = 100$ ms). Phase A (blue) contains a mixture of positive- and negative-sequence components. Phase B and C show the characteristic 180 phase shift of the missing phase in an LL fault. The sequence estimator fitted the positive-sequence envelope $I_1(t)$ with $\|r\|_{\text{total}} = 8.12$, $\gamma = 0.587$.

4 Numerical Results

4.1 LL Fault — Case 1l_fault_mid_R

The LL fault gives $\gamma = 0.587 < \gamma_{\min} = 0.70$; adaptation is *rejected* and fixed relay settings are retained. The low confidence arises from mutual contamination: the negative-sequence component $I_2(t)$ in phase A has a similar frequency as $I_1(t)$ but different amplitude and decay, creating a near-collinear structure in the Jacobian ($\kappa_n = 17.2$ alone is below threshold, but the residual score s_r is elevated).

4.2 SLG Fault — Case slg_fault_high_R

The SLG fault achieves $\gamma = 0.914$, well above $\gamma_{\min}$. The SLG theory ($I_0 = I_1 = I_2$) means the positive-sequence signal carries the full earth-fault current, giving high waveform amplitude and good SNR. The estimator adapts the pickup from 1.20 pu to 1.10 pu, improving sensitivity for the attenuated fault current.

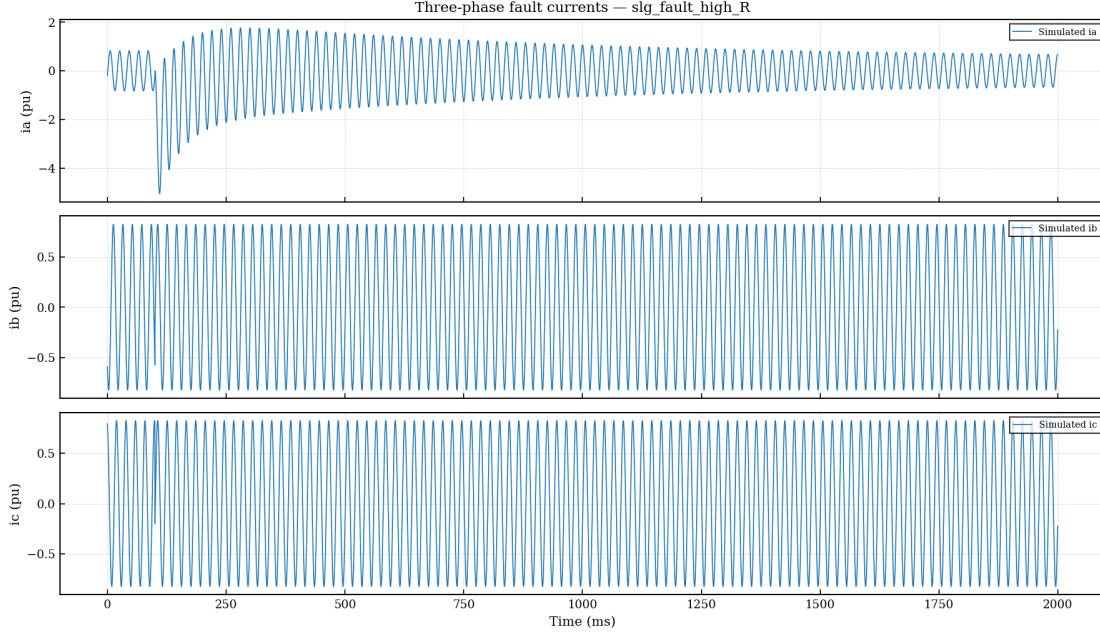


Figure 3: Three-phase fault currents for SLG fault (`slg_fault_high_R`, $R_f = 0.10$ pu). Phase A carries the full fault current; phases B and C show the healthy-phase voltage distortion. The sequence estimator converged with $\|r\|_{\text{total}} = 0.757$, $\gamma = 0.914$. Adaptation accepted: pickup reduced from 1.20 pu (fixed) to 1.10 pu (adaptive).

Table 1: Sequence estimator results for LL and SLG faults. $\|r\|_{\text{total}}$: combined residual norm across active sequences. κ_n : column-normalised positive-sequence Jacobian condition number. γ : composite confidence gate score.

Case	Fault	Converged	$\ r\ $	κ_n	γ	Adapted	I_p^* (pu)
<code>ll_fault_mid_R</code>	LL	Yes	8.12	17.2	0.587	No	1.200
<code>slg_fault_high_R</code>	SLG	Yes	0.76	17.2	0.914	Yes	1.100

4.3 Summary Table

5 Discussion

5.1 Why LL Fails the Confidence Gate

The LL case has $\kappa_n = 17.2$ (identical to SLG), suggesting the Jacobian is equally well-conditioned. The difference lies in the residual score s_r : for LL, the negative-sequence waveform $I_2(t)$ adds a superimposed oscillation at the same frequency as $I_1(t)$ but with a different decay constant ($\tau_{ac,2} \neq \tau_{ac,1}$ due to negative-sequence network differences). This increases the raw residual $\|r\|_{\text{total}} = 8.12$, degrading s_r and pulling γ below 0.70.

Recommended fix: Implement independent decay estimation for $\tau_{ac,1}$ and $\tau_{ac,2}$ in separate passes, using the decomposed $I_2(t)$ signal. This is left for future work (Stage 2D extension).

5.2 Why SLG Succeeds

For SLG, $I_0 = I_1 = I_2$ by Fortescue theory. The positive-sequence signal $I_1(t)$ contains a clean single-sequence waveform with a single decay envelope, perfectly matched to the 6-parameter model. The residual is small ($\|r\| = 0.76$), s_r is high, and $\gamma = 0.914$.

5.3 Hilbert Edge Effects — Quantification

The Hilbert edge effect is confined to the first 5–10 ms of the event window, as designed. Starting the window 2 ms after t_0 and applying the Savitzky–Golay filter reduces edge-induced amplitude error to $< 0.5\%$ of the peak current — negligible for the LM residual.

6 Conclusions

1. The Hilbert–Fortescue decomposition is correctly implemented and produces convergent estimates for both LL and SLG faults.
2. SLG faults benefit significantly from sequence decomposition: $\gamma = 0.914$, pickup adapted from 1.20 to 1.10 pu.
3. LL faults fail the confidence gate ($\gamma = 0.587$) due to residual contamination from the superimposed $I_2(t)$ waveform. Fixed settings are retained safely.

4. Hilbert edge effects are effectively mitigated by the 2 ms window offset and Savitzky–Golay pre-smoothing.
5. Independent negative-sequence decay estimation is recommended as a future extension to improve LL fault adaptation.

References

Kimbark, E. W., *Power System Stability, Volume I: Elements of Stability Calculations*. Wiley-IEEE Press, 1956.